

Zachary James

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I use Generative AI (diffusion models), Gaussian Processes, and High-Performance Computing to analyze spatial data.

Education

Cornell University, College of Computing

Ph.D. in Statistics and Data Science, Minor in Artificial Intelligence, GPA: 3.9/4.0

Ithaca, NY
December 2026

Georgia Institute of Technology, College of Computing

Bachelor of Science in Computer Science, GPA: 4.0/4.0

Atlanta, GA
May 2021

Research Areas: Gaussian Processes, Diffusion Models, Geospatial AI

Work Experience

Cornell Bowers College of Computing

Graduate Researcher

Ithaca, NY
August 2022- Present

- Using image segmentation and conditional diffusion models to explain relationships in climate data
- Building GPU-enabled software for interpolating satellite imagery at scale

New York City Government

Research Fellow

New York, NY
May 2023-August 2023

- Created a Gaussian process model in R to increase the resolution of weather forecasts
- Hosted a dashboard with model output on AWS EC2 instance for NYC government use

Amazon

Software Engineering Intern

Seattle, WA
May 2020-August 2020

- Built a new Tier 1 two-factor authentication system used in production by millions of users
- The new product relied entirely on services from AWS, such as DynamoDB and Pinpoint, and allowed Amazon to deprecate an older service

Cary Institute of Ecosystem Studies

Data Science Intern

Millbrook, NY
May 2019-August 2019

- Developed a spatial CNN image classifier in Python with TensorFlow and OpenCV to identify fruit bats. Achieved 91% accuracy on a balanced dataset
- Created R scripts to automate the preprocessing of forestry data for statistical analysis. Conducted ANOVA and produced data visualizations to help ecologists understand the relationship between biomass and soil nutrients.

Research + Software

A Framework for Nonstationary Gaussian Processes with Neural Network Parameters [\[Paper\]](#) [\[Github\]](#)

[Z James](#), J Guinness, Under Review

- Novel interpretable nonstationary GP model combines Bayesian inference with neural nets for pixel interpolation
- Achieves up to 75% lower MSE than existing models on benchmark datasets
- Open source PyTorch + GPyTorch implementation available with feed-forward and convolutional neural nets

Implementation and analysis of GPU algorithms for Vecchia Approximation [\[Paper\]](#) [\[Github\]](#)

[Z James](#), J Guinness, Statistics and Computing 34 (6), 207, 2024 (Acceptance rate < 25%)

- R package built with CUDA and C++ for fast large-scale Gaussian Process modeling on NVIDIA GPUs.
- My novel algorithm trains models 20 times faster than CPU software, 7 times faster than existing GPU software

Doctor Shopping and the Patient Sharing Network of Healthcare Providers [\[Paper\]](#)

[Z James](#), N Serban, 2021, Preprint

- Constructed a random graph model with 22k node and 1 million edges to better understand opioid addiction
- Analyzed graph in Python and R to discover urbanity is largest determinant of opioid abuse

Technical Skills

- **Machine Learning Tools:** Pytorch, Stan
- **High Performance Computing:** CUDA/C++ Development, OpenMP, R, GitHub

Teaching and Service

Head TA: Python Programming and its Applications in Statistics

TA: Probability Models and Inference, Statistical Methods I

Service: Cornell Math Department Directed Reading Program (Undergraduate Mentorship)